TABLE VI: **Ablation Study of Object Head Design on RoboTwin 2.0 tasks.** We compare two strategies for supervising attention heads in the Object Head: region-based supervision using binary masks from foundation models, and Gaussian prior–based KL divergence. The region-based method leads to significantly higher performance (83.33% average success rate), especially on precision-critical tasks such as *Click Bell*, confirming the advantage of providing explicit spatial constraints over soft priors when grounding object-level attention.

Model	Beat Hammer Block	Click Bell	Dump Bin BigBin	Avg
π_0 w/ gaussian	89%	40%	87%	72.00%
π_0 w/ object region **(Ours)**	**94%**	**62%**	**94%**	**83.33%**

TABLE VII: **Ablation Study of Depth Head Design on RoboTwin 2.0 tasks.** We evaluate the impact of encoder capacity and token downsampling in our Depth Head. The best results are achieved using the `Depth Anything 3-small` variant (83.00% average success rate), which balances compactness and representational power. Larger encoders offer no significant benefit and may introduce redundancy. Removing token downsampling leads to a notable drop in performance (68.00%), especially on fine-grained tasks like *Click Bell*, supporting the need for moderate spatial abstraction in depth-guided attention.

Model	Beat Hammer Block	Click Bell	Dump Bin BigBin	Avg
π_0 w/ small encoder **(Ours)**	**96%**	<u>63%</u>	**87%**	**83.00%**
π_0 w/ base encoder	77%	49%	<u>83%</u>	69.67%
π_0 w/ large encoder	<u>92%</u>	**67%**	**87%**	<u>82.00%</u>
π_0 w/ small encoder w/o downsample	78%	39%	**87%**	68.00%

TABLE VIII: **Ablation Study on Skill Head Supervision.** We compare Skill Head variants trained with hard vs. soft label supervision on RoboTwin 2.0 tasks. The soft label variant (ours) achieves the highest average performance (75.00%), showing improved generalization on less structured tasks such as *Click Bell* and *Dump Bin BigBin*.

Model	Beat Hammer Block	Click Bell	Dump Bin BigBin	Avg
π_0 w/ hard labels	90%	36%	82%	69.33%
π_0 w/ soft labels **(Ours)**	**92%**	**39%**	**94%**	**75.00%**

gies: direct addition, gated modulation, and zero-initialized convolution (ours). These strategies modulate the main attention stream with external signals in different ways, balancing simplicity, control, and stability.

Direct Addition. A straightforward approach is to directly add the auxiliary attention features $\mathrm{Attn}_{\text{specified}}(\mathbf{x})$ to the main attention stream $\mathrm{Attn}_{\text{main}}(\mathbf{x})$:

$$\mathrm{Attn}(\mathbf{x}) = \mathrm{Attn}_{\text{main}}(\mathbf{x}) + \mathrm{Attn}_{\text{specified}}(\mathbf{x}). \tag{17}$$

While simple and computationally efficient, this method lacks any adaptive control over the fused signal. It may lead to feature interference or training instability, particularly in tasks requiring fine-grained control.

Gated Addition. To enable learnable modulation, we apply a nonlinear transformation to the auxiliary features using a $\tanh$ gate before fusion. This produces the final attention as:

$$\mathrm{Attn}(\mathbf{x}) = \mathrm{Attn}_{\text{main}}(\mathbf{x}) + \tanh\left(\mathrm{Attn}_{\text{specified}}(\mathbf{x})\right). \tag{18}$$

Unlike scalar gating, this formulation allows each element of the auxiliary feature map to be adaptively scaled within $(-1, 1)$, enabling smoother gradients and more expressive control. However, the unbounded nature of the fused signal can still introduce training instability in tasks requiring fine spatial precision.

Zero-initialized Convolution. Inspired by residual modulation in ControlNet, we propose applying a zero-initialized convolutional layer to the auxiliary features before fusion. This yields the final output as:

$$\mathrm{Attn}(\mathbf{x}) = \mathrm{Attn}_{\text{main}}(\mathbf{x}) + \mathrm{ZeroConv}\left(\mathrm{Attn}_{\text{specified}}(\mathbf{x})\right). \tag{19}$$

The zero initialization ensures that the fused path initially behaves as an identity function, preventing early training collapse. The network can then gradually learn to utilize auxiliary cues in a stable and interpretable manner. Empirically, this design leads to more consistent performance gains across a range of tasks.

Experiments and Results. We evaluate all three fusion methods on the *RoboTwin 2.0* benchmark. As summarized in Table IX, our zero-initialized convolution strategy achieves the